Online

Advanced Network Administration Using Linux

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Terminal

To start a terminal use (ctrl+alt+T)

Commands

• Sudo → executes a command with admin privileges (root-user)

File and Directory Commands

• Tilde (~) stands for home directory
Ex: If you are 'user', tilde (~) stands for "/home/user".

• pwd (print Working Directory) allows you to know which directory you're located.
Ex: pwd in the 'desktop' directory will show "~/Desktop".

• ls (list) will show/list files in the current directory.
Ex: ls in home, ls~, will show files in the home directory.

• cd allows you to change directories.
When opening a terminal you will be in the home directory. To move around you will use cd. To go to root, use (cd /), to home (cd or cd ~), up one level (cd ..), to previous directory, or back (cd -), for a specific directory (cd ~/Desktop).

• cp (copy) allows you to copy a file. To copy a directory use "cp -r directory foo" (copy -recursivery).
EX: "cp file foo" makes copy of 'file' named "foo"

• mv (move) allows you to move a file to a different location or will rename a file.
EX: "mv file foo" renames "file" to "foo"
"mv foo ~/Desktop" will move the file "foo" to the Desktop directory.

• rm (remove) allows you to remove or delete a file in your directory.

• rmdir (remove directory) will delete an empty directory. To delete a directory and all of its contents recursivery use "rm -r" instead.

• mkdir (make directory) allows you to create directories.

Main Directories

The standard Ubuntu directory structure mostly follows the Filesystem Heirarchy Standard. These are the most important directories

• /bin a place for most commonly used terminal commands.

• /boot contains files needed to start up the system, including the linux kernel, a RAM disk image, and bootloader configuration files.

• /dev contains all device files, which are not regular files but instead refer to various hardware devices on the system, including harddrives.

• /etc contains system-global config files, which affect the system's behavior for all users.

• /home place for user's home directory.

• /lib contains very important dynamic libraries and kernel modules.

• /media is intended as a mount point for external devices, such as hard drives or removeable media (Floppies, CDs, DVDs).

• /mnt is also a place for mount points, but is dedicated specifically to "temporarily mounted" devices, such as network filesystems

• /opt can be used to store additional software for your system, which is not handled by the package manager

- /proc is a virtual filesystem that provides a mechanism for kernel to send information to processes.

- /root is the superuser's home directory, not in /home to allow for booting the system even if /home is unavailable.

- /run is a tmpfs (temporary filesystem) available early in the boot process where ephemeral data is stored. Files under this directory are removed or truncated at the beginning of the boot process.

- /sbin contains important admin commands that should generally only by employed by the superuser.

- /srv can contain data directories of services such as HTTP (/srv/www/) or FTP.

- /sys is a virtual filesystem that can be accessed to set or obtain information about the kernel's view of the system.

- /tmp is a place for temporary files used by applications.

- /usr contains the majority of most user utilities and applications, and partly replicates the root directory structure, containing for instance, among others, /usr/bin and /usr/lib.

- /var is dedicated to variable data, such as logs, databases, websites, and temporary spool (email, etc.) files that persist from one boot to the next. A notable directory it contains is /var/log where system log files are kept.